Planar Drone: Dynamics and Control

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CDS 232: Nonlinear Dynamics
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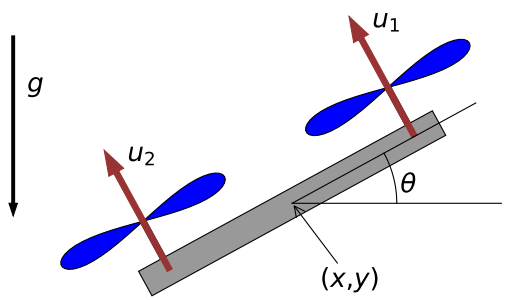
I. System Overview

Autonomous aircraft have demonstrated unparalleled versatility and scalability for distributed tasks in areas such as agriculture, surveillance, delivery, surveying and more. In this project, we study a quadcopter modeled as a rigid body with three degrees of freedom, subject to thrust and moment inputs generated by four independently actuated rotors. The following document details a derivation of the vehicle dynamics, describing its translation and orientation.



3 DoF Planar Drone

One of the research goals of this system is to develop controllers that ensure safety guarantees under disturbance uncertainty. For instance, consider a drone used for aerial package delivery. As the drone travels, the weight of the packages will act as a swinging pendulum, disturbing the natural flight of the aircraft. This disturbance will inherently be influenced by the weight and the geometry of the payload. How can we design a controller robust to this uncertainty?

II. System Dynamics

For a drone of mass m and thrust inputs u_1, u_2, the translational dynamics expressed in the inertial frame are given by

$$m\ddot{x} = -(u_1 + u_2)\sin\theta$$

$$m\ddot{y} = (u_1 + u_2)\cos\theta - mg$$

The rotational dynamics are given by

$$I\ddot{\theta} = r(u_1 - u_2)$$

where I is the moment and r is the distance from the motors to the CoM.

Considering the state vector

$$x = \begin{bmatrix} x & \dot{x} & y & \dot{y} & \theta & \dot{\theta} \end{bmatrix}$$

We write the system dynamics as follows

$$\dot{x} = \begin{bmatrix} \dot{x} \\ -(u_1 + u_2)\sin\theta/m \\ \dot{y} \\ (u_1 + u_2)\cos\theta/m - g \\ \dot{\theta} \\ r(u_1 - u_2)/I \end{bmatrix}$$

III. Baseline Controller

Given a desired state and control input along the trajectory (x_{ref}, u_{ref}), define the errors

$$e = x - x_{ref}$$

$$e_u = u - u_{ref}$$

such that the error dynamics are

$$\begin{aligned} \dot{e} &= \dot{x} - \dot{x}_{ref} \\ &= f(x, u) - f(x_{ref}, u_{ref}) \\ &= f(x_{ref} + e, u_{ref} + e_u) - f(x_{ref}, u_{ref}) \end{aligned}$$

Let the function

$$g(e, e_u) = f(x_{ref} + e, u_{ref} + e_u) - f(x_{ref}, u_{ref})$$

Linearize the error dynamics about $(0, 0)$

$$g(e, e_u) \approx g(0,0) + \left.\frac{\partial g}{\partial \tilde{x}}\right|_{0,0} e + \left.\frac{\partial g}{\partial e_u}\right|_{0,0} e_u$$

Note $g(0,0) = 0$. By the chain rule we have

$$\left.\frac{\partial g}{\partial \tilde{x}}\right|_{0,0} = \left.\frac{\partial f}{\partial x}\right|_{x_{ref}, u_{ref}} \qquad \left.\frac{\partial g}{\partial e_u}\right|_{0,0} = \left.\frac{\partial f}{\partial u}\right|_{x_{ref}, u_{ref}}$$

Thus, we are left with the linear system

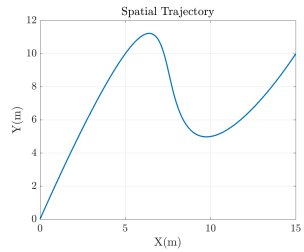
$$\dot{e} = Ae + Be_u$$

The explicit form of A and B are presented in the index.

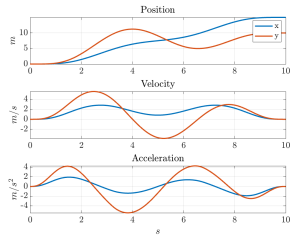
Finally, consider the control law of the form $u = u_{ref} - ke$ where the gain k is computed as LQR.

IV. Trajectory Generation

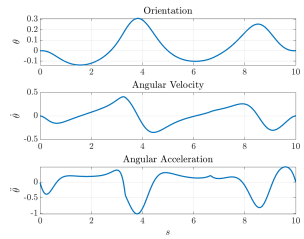
The nominal trajectory (x_{ref}, u_{ref}) is generated by solving a Minimum Snap Optimization Problem. Consider the following reference trajectory



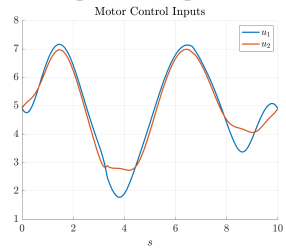
(a) Spatial Trajectory



(b) Spatial Components

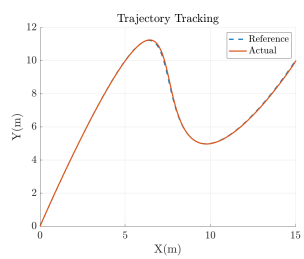


(c) Orientation

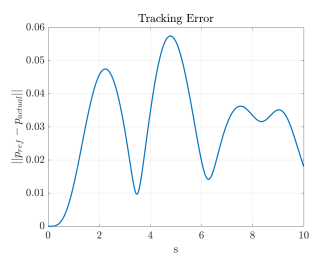


(d) Control Inputs

Implementing the trajectory tracking controller described above we observe the following performance



(a) Spatial Trajectory



(b) Spatial Components

Note that the state and control cost can be tuned in the LQR formulation for improved trajectory tracking.

V. Lipschitz Continuity

To show the nonlinear dynamics $\dot{x} = f(x)$ are Lipschitz continuous, consider taking the Jacobian

$$Df(x) = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & A_{2,5} & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & A_{4,5} & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

where

$$A_{2,5} = -\frac{(u_1 + u_2)}{m} \cos\theta$$

$$A_{4,5} = -\frac{(u_1 + u_2)}{m} \sin\theta$$

Given that f is continuously differentiable, we can conclude it is locally Lipschitz. Thus, we can find a Lipschitz constant such that

$$||Df(x)|| \leq L$$

Given the Lipschitz continuity condition must hold for any norm $||\cdot||$, we take the 2-norm as follows. Let $A = Df(x)$

$$||A||_2 = \sqrt{\lambda_{max}(A^\top A)}$$

Note that

$$A^T A = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{(u_1+u_2)^2}{m^2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Given that $A^\top A$ is a diagonal matrix, eigenvalues are the diagonal elements. Hence we have

$$||A||_2 = \sqrt{\max\left(1, \frac{(u_1 + u_2)^2}{m^2}\right)}$$

Consider the motor limits u_{max} such that $u_1, u_2 \leq u_{max}$. Evaluating at these maximum bounds provides a global Lipschitz constant for the specified domain of inputs:

$$\begin{aligned} L = ||A||_2 &= \sqrt{\frac{(2u_{max})^2}{m^2}} \\ &= \frac{2u_{max}}{m} \end{aligned}$$

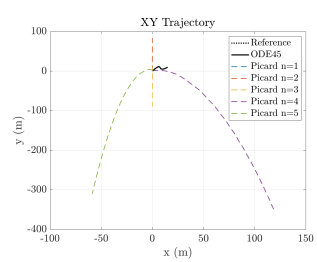
VI. Solutions Through Approximations

To compute solutions numerically, we employ Picard Iteration using the recursive integral formula

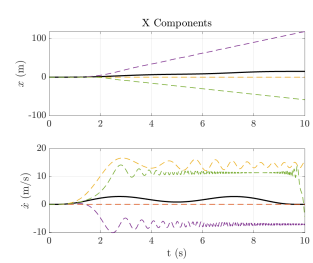
$$\phi_0(t) = x_0$$

$$\phi_{k+1}(t) = x_0 + \int_{t_0}^{t} f(\phi_k(s))ds$$

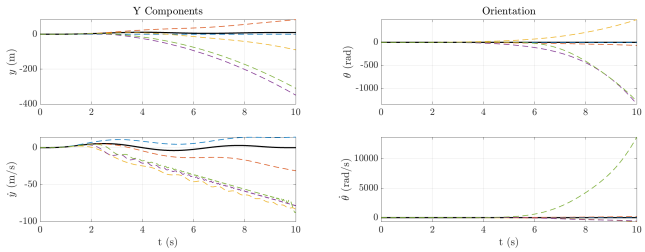
The following figures compare the converged Picard solutions against a standard Runge-Kutta (RK45) numerical solver to verify the accuracy of the approximation.



(a) Spatial Trajectory



(b) X Components



(a) Y Components (b) Orientation

VII. Dependence on Parameters and Initial Conditions

Consider the perturbed system with solution $z(t)$ to the following IVP

$$\dot{z} = f(z) + g(z) \qquad z(t_0) = z_0$$

We construct $g(z)$ such that the mass of the drone is doubled, representing a parameter perturbation. We also introduce a perturbation in the initial condition z_0.

$$g(z) = \begin{bmatrix} 0 \\ (u_1 + u_2)\sin\theta/2m \\ 0 \\ -(u_1 + u_2)\cos\theta/2m \\ 0 \\ 0 \end{bmatrix}$$

with the resulting dynamics

$$\dot{z} = \begin{bmatrix} \dot{x} \\ -(u_1 + u_2)\sin\theta/2m \\ \dot{y} \\ (u_1 + u_2)\cos\theta/2m - g \\ \dot{\theta} \\ r(u_1 - u_2)/I \end{bmatrix}$$

From this, we can compute a μ used to exponentially bound the growth error such that $||g(z)|| \leq \mu$. Taking the 2-norm:

$$\begin{aligned} \mu &\geq \sqrt{\frac{(u_1 + u_2)^2\cos^2\theta}{4m^2} + \frac{(u_1 + u_2)^2\sin^2\theta}{4m^2}} \\ &\geq \frac{u_1 + u_2}{2m} \end{aligned}$$

For $u_1 = u_2 = u_{max}$, we find $\mu = 10$. By Theorem 3.2 we can construct a bound on the error growth between the nominal solution $x(t)$ and the perturbed solution $z(t)$:

$$||x(t) - z(t)|| \leq ||x_0 - z_0||e^{L(t-t_0)} + \frac{\mu}{L}(e^{L(t-t_0)} - 1)$$

The plot below illustrates that the actual norm of the solution difference remains well below this theoretical exponential bound, confirming the continuous dependence on initial conditions and parameters.

For the initial conditions

$$x_0 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$
$$z_0 = \begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 0 \end{bmatrix}$$

we have the error growth

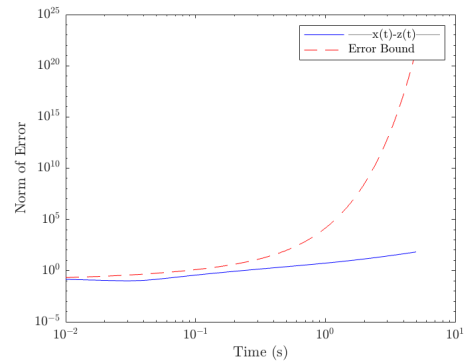


Fig. 5: Exponential Bound on Error Growth

VIII. Hartman-Grobman Theorem

We verify the topological equivalence of the linearization Consider the control law of the form $u = u_0 - k\tilde{x}$ where the gain k is computed as LQR. We are then left with the autonomous system

$$\begin{aligned} \dot{\tilde{x}} &= (A - Bk)\tilde{x} \\ &= A_{aug}\,\tilde{x} \end{aligned}$$

We verify the local equivalence of the linearization to the nonlinear system through Hartman-Grobman.

$$\lambda(A) = \begin{bmatrix} -1.0111 + 1.5707i \\ -1.0111 - 1.5707i \\ -1.9667 \\ -1.0108 \\ -1.0987 + 0.4551i \\ -1.0987 - 0.4551i \end{bmatrix}$$

Given that all eigenvalues have a nonzero real part, we can conclude $(0, 0)$ is a hyperbolic equilibrium of the error dynamics.

IX. Lyapunov Stability

Consider the candidate Lyapunov function

$$V(x) = x^\top P x$$

where $P = P^\top \succ 0$ satisfying the CTLE

$$A^\top P + PA = -Q$$

From the conditions for stability we have

$$\lambda_{min}(P)\|x\|^2 \leq V(x) \leq \lambda_{max}(P)\|x\|^2$$

We verify the second condition we have

$$\begin{aligned}\dot{V}(x) &= \dot{x}^\top P x + x^\top P \dot{x} \\ &= x^\top A^\top P x + x^\top P A x \\ &= x^\top (A^\top P + PA) x \\ &= -x^\top Q x\end{aligned}$$

Thus, we have

$$\dot{V}(x) \leq -\lambda_{min}(Q)\|x\|^2$$

In this case, taking $Q = I_6$ we compute P for the closed loop system, which yields the eigenvalues

$$\lambda(P) = \begin{bmatrix} 0.1129 \\ 0.2701 \\ 1.1506 \\ 1.4446 \\ 6.7640 \\ 7.7801 \end{bmatrix}$$

Thus, we have

$$0.1129\ \|x\|^2 \leq V(x) \leq 7.7801\ \|x\|^2$$

$$\dot{V}(x) \leq -\|x\|^2$$

X. Region of Attraction

We provide an inner approximation of the region of attraction using the constructive method of Theorem 8.3. By the mean value theorem, the closed-loop nonlinear dynamics admit the decomposition

$$\dot{e} = A_{cl} e + G(e) e$$

$$G(e) = \int_0^1 Df_{cl}(\tau e)\, d\tau - Df_{cl}(0)$$

where Df_{cl} is the Jacobian of the closed-loop dynamics $f_{cl}(e) = f(x_0 + e,\, u_0 - Ke)$. By Theorem 8.3, exponential stability holds on the ball $B_r(0)$ provided

$$N \triangleq \max_{\|e\| \leq r} \left\| \int_0^1 Df_{cl}(\tau e)\, d\tau - Df_{cl}(0) \right\| < \frac{\lambda_{\min}(Q)}{2\lambda_{\max}(P)}$$

We compute N by solving the constrained optimization problem

$$\max_e \quad \|G(e)\| \quad \text{subject to} \quad \|e\|^2 \leq r^2$$

where the integral is evaluated via approximated numerically and the maximization is performed using multi-start `fmincon`. A bisection search over r yields the critical radius r^* at which the bound is saturated. With $Q = I_6$, we obtain

$$\frac{\lambda_{\min}(Q)}{2\lambda_{\max}(P)} = \frac{1}{2\lambda_{\max}(P)}$$

We define the inner approximation of the Region of Attraction as the ellipsoid

$$\Omega_\beta = \{e \in \mathbb{R}^6 : e^\top P e \leq \beta\}, \quad \beta = \lambda_{\min}(P)\, r^{*2}$$

Implementing the program describe above we obtain

$$r^* \approx 5 \times 10^{-3}$$

XI. Barrier Functions for Obstacle Avoidance

Consider the barrier function candidate

$$h(x) = \|x - x_{obs}\| - D_{obs}$$

We seek to enforce the condition

$$\dot{h}(x, v) = \frac{(x - x_{obs})^\top}{\|x - x_{obs}\|} v \geq \alpha h(x)$$

Note that there are no input terms in this expression, i.e. we say h has relative degree 2. Thus, we construct the extended barrier function

$$\begin{aligned}h_e(x, v) &= \dot{h}(x, v) + \gamma_1 h(x) \\ &= \frac{(x - x_{obs})^\top}{\|x - x_{obs}\|} v + \gamma_1 \|x - x_{obs}\| - \gamma_1 D_{obs}\end{aligned}$$

Taking a derivative along

$$\begin{aligned}\dot{h}_e(x, v, a) = &\frac{v^\top v + (x - x_{obs})^\top a}{\|x - x_{obs}\|} \\ &- \frac{((x - x_{obs})^\top v)^2}{\|x - x_{obs}\|^3} + \gamma_1 \frac{(x - x_{obs})^\top v}{\|x - x_{obs}\|}\end{aligned} \tag{1}$$

where $a = \begin{bmatrix} \ddot{x} \\ \ddot{y} \end{bmatrix} = \begin{bmatrix} -(u_1 + u_2)\sin\theta/m \\ (u_1 + u_2)\cos\theta/m - g \end{bmatrix}$

We seek to enforce the condition

$$\dot{h}_e(x, v, a) \geq \gamma_2 h(x, v)$$

In order to blend the barrier-safety controller with the nominal trajectory following controller, we seek to find the optimal u

$$u^* = \min_u \frac{1}{2}\|u_{nom} - u\|^2$$

subject to the constraint

$$\dot{h}_e(x, v, a) \geq -\gamma_2 h(x, v)$$

Employing this CBF on a simple trajectory, we observe

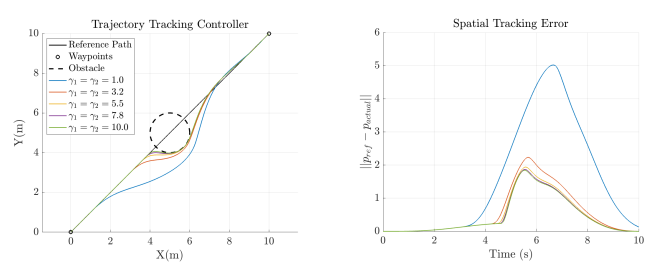


Fig. 6: Barrier function across different values of γ_1, γ_2

The controller successfully pushes the drone away from the obstacle. Increasing values of gamma weaken the safety condition, allowing the drone to come close to the obstacle.

Applying the CBF to the trajectory shown earlier, we can observe the following performance.

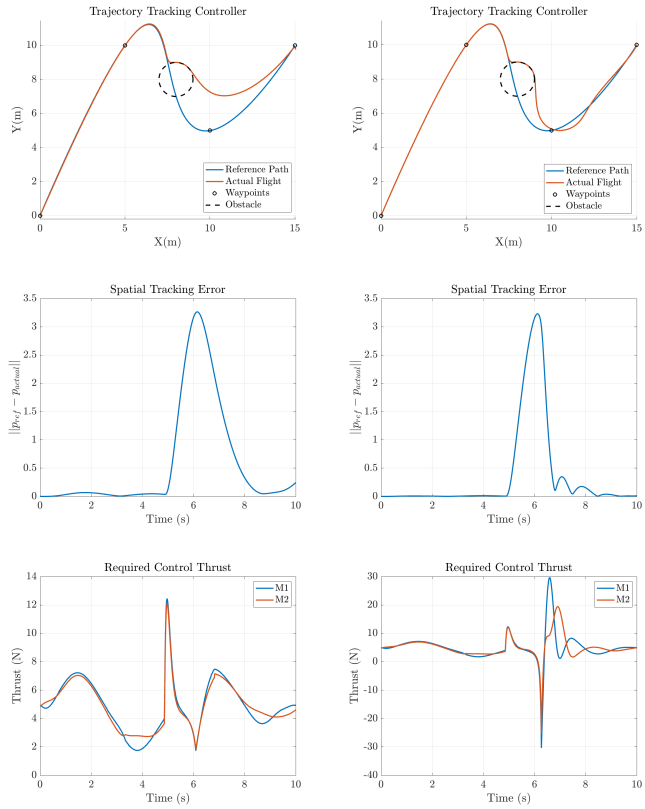


Fig. 7: Comparison of low cost tracking controller (left) vs aggressive tracking controller (right)

We can observe that as the controller pushes the drone away from the obstacle, it begins to fall behind along the desired trajectory. This could be compensates by increasing the state cost in the LQR formulation (shown on the right). Although this leads to tighter tracking, it blows up the desired control inputs and overshoots, leading to oscillations along the remainder of the trajectory.

XII. Future Work

We have shown the ability of constructing barrier functions to render a safe set in the presence of a trajectory tracking controller that appears to violate safety conditions. However, the by blending these two control inputs, we give up the ability to follow the trajectory ”on-time”. Therefore, a future research direction of this project is to explore methods for online planning in the presence of safety conditions. For instance, solving an MPC formulation with a CBF layer. Futhermore, during the derivation above, it was assumed that the obstacle was at a fixed location. Thus, another interesting research direction is to formulate CBFs for obstacle avoidance of moving targets.